

MINspire 2026 Capstone Project

Predicting House Prices in the Kathmandu Valley

An Exploratory Data Analysis & Regression Study of the Nepali Housing Market

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1 Problem Statement

Buying a house is the single largest financial decision most Nepali families will ever make, yet the market is remarkably opaque. Listed prices are set by individual sellers and brokers with no public benchmark, so a buyer has little objective way to judge whether an asking price is fair. This project asks a simple but practically useful question:

Can we predict the listed price of a house in the Kathmandu Valley from its observable characteristics, land area, number of bedrooms and bathrooms, parking, road access and location — and which of these factors matter most?

We treat this as a supervised regression problem, in the tradition of *hedonic* pricing models that decompose the value of a good into the implicit prices of its attributes (Rosen, 1974; Harrison and Rubinfeld, 1978). The goals are twofold: (1) build a model that estimates a fair market price for a property, and (2) interpret the model to understand the principal drivers of price in the valley. A reliable answer would help buyers sanity-check listings, help sellers price competitively, and quantify intuitions (“land area matters most”) that are usually only assumed.

2 Dataset Description

The data is the publicly available *Nepali Housing Price Dataset* (Thapa, 2020), a snapshot of online property listings scraped on 27 April 2020. The raw file contains **2,211 listings described by 18 columns**. Each row is a single advertised property. The key fields are summarised in Table 1.

Table 1: Key columns in the raw dataset.

Column	Description	Type
Price	Listed price in Nepali Rupees (NPR), target variable	Numeric
Area	Land area, written in mixed units (Aana, Ropani, sq.ft, etc.)	Text
Bedroom / Bathroom	Room counts	Numeric
Floors	Number of storeys (often blank)	Numeric
Parking	Parking capacity (vehicles)	Numeric
Face	Compass direction the house faces	Category
City / Address	Location of the property	Text
Road Width / Type	Width and surface of the access road	Text
Amenities	List of features (garden, lift, etc.)	Text (list)
Year, Views, Posted	Build year, listing views, time posted	Mixed

A note on units. Nepal uses two traditional land-measurement systems: the Ropani–Aana–Paisa–Daam system in the hills and Kathmandu Valley, and the Bigha–Kattha–Dhur system in the Terai plains. In this dataset land area appears in nine different unit labels, frequently encoded as a four-part string such as “1-0-0-0 Aana”. Converting every record to a single unit (square feet) was the central data-engineering challenge of the projects the conversion factors used were 1 Ropani = 5,476 sq.ft, 1 Aana = 342.25 sq.ft, 1 Paisa = 85.56 sq.ft and 1 Daam = 21.39 sq.ft.

3 Data Cleaning and Preparation

Real listing data is noisy, and this dataset is no exception. Before any modelling we performed an extensive cleaning and scoping process. The most important decisions were:

- Parsing land area into a single numeric column (square feet) using a custom parser that interprets the four-part Ropani–Aana strings and direct units, and discards unparseable entries such as “dont know Sq. Feet”.
- Removing junk prices. Listed prices ranged from NPR 15 to NPR 216 trillion, clearly placeholder or mistyped values. We restricted to a realistic window of NPR 1 million to 300 million.
- Scoping to the Kathmandu Valley (Kathmandu, Lalitpur, Bhaktapur), which contains $\approx 90\%$ of listings and shares a single land-measurement system, so the model is internally consistent rather than mixing incompatible regional markets.
- Keeping only listings with at least one bedroom. Roughly half the rows had Bedroom = 0 these are bare-land plots and incomplete listings rather than houses, so they were excluded from a house-price model.
- Imputing missing Floors and Road Width with the median, treating missing Road Type as a separate “Unknown” category, and dropping the Year column entirely (74% missing and mixing the Bikram Sambat and Gregorian calendars).
- Feature engineering: counting amenities per listing, and converting all road widths to feet.

The effect of each step on the sample size is shown in Table 2. The final modelling set is a clean, well-defined population: actual houses in the Kathmandu Valley with a plausible price and a parseable land area.

Table 2: *Cleaning funnel, listings remaining after each step.*

Cleaning step	Listings remaining
Raw listings	2,211
Kathmandu Valley only	2,000
Actual houses (Bedroom ≥ 1)	1,054
Parseable land area	1,031
Price within NPR 1M – 300M	792
Plausible area (100 – 20,000 sq.ft)	738

4 Exploratory Data Analysis and Visualisations

4.1 The shape of price

House prices are extremely right-skewed: most homes cluster at the low end with a long tail of luxury properties. Taking the base-10 logarithm of price produces an approximately symmetric, bell-shaped distribution (Figure 1). This is the single most important EDA finding, it tells us to model $\log_{10}(\text{Price})$ rather than raw price, which both stabilises the variance and prevents a handful of mansions from dominating the fit.

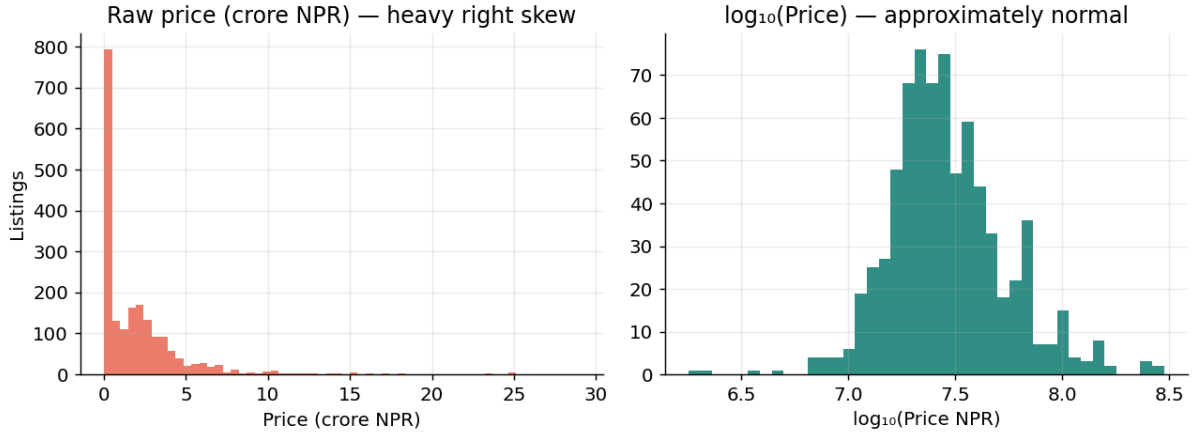


Figure 1: *Raw price is heavily right-skewed (left) $\log_{10}(\text{price})$ is approximately normal (right).*

4.2 Where the listings are

The market is overwhelmingly concentrated in the Kathmandu Valley, which justifies our decision to scope the model there (Figure 2). Within the valley, Kathmandu commands the highest median price, followed by Bhaktapur and Lalitpur (Figure 3).

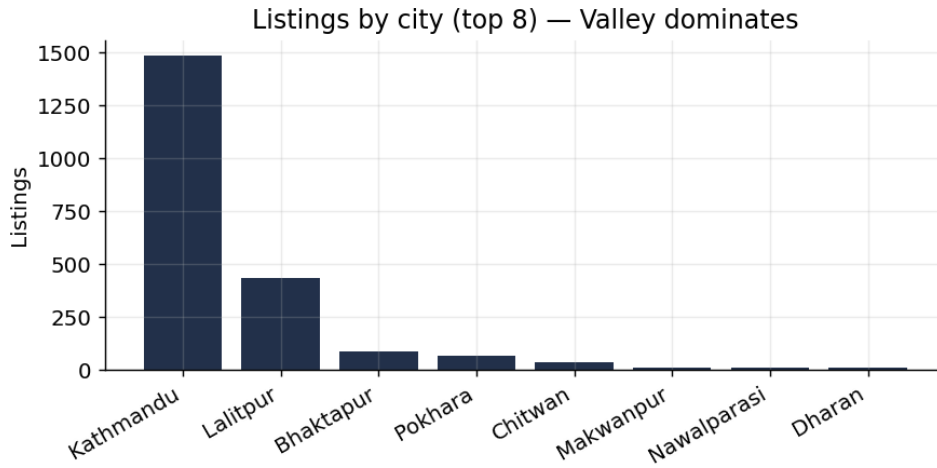


Figure 2: *Listing counts by city, the Kathmandu Valley dominates the dataset.*

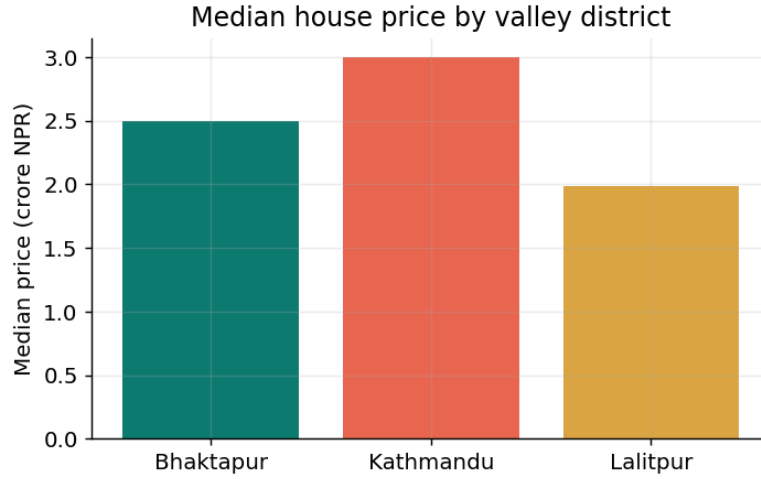


Figure 3: Median house price by valley district (Kathmandu $\approx$ NPR 3.0 crore).

4.3 Missing data

Several columns are substantially incomplete, which directly shaped our cleaning strategy (Section 3). Year and Floors are missing for the majority of rows (Figure 4).

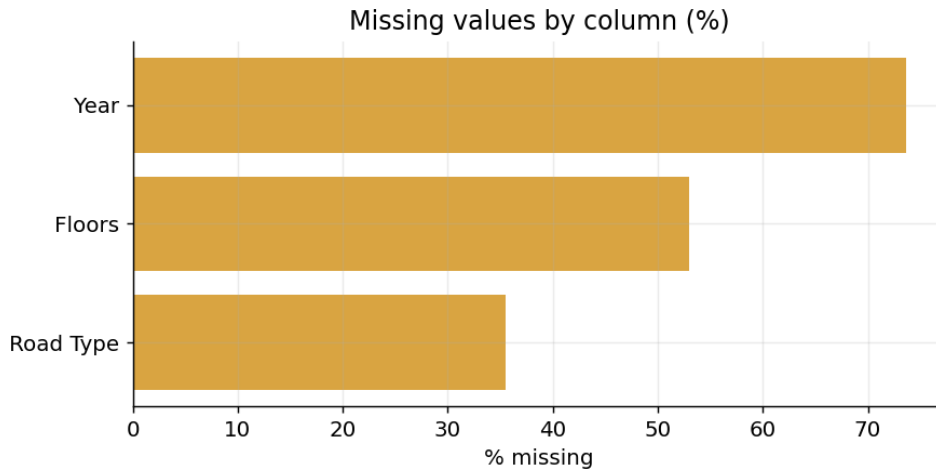


Figure 4: Percentage of missing values by column.

4.4 What drives price

Land area is the clearest single predictor of price. On a log-log scale the relationship is positive and roughly linear, although the scatter is wide, area alone explains only part of the story (correlation $r \approx 0.31$, Figure 5). The correlation matrix (Figure 6) confirms that area, bathrooms and parking are the variables most associated with price, while bedroom count is surprisingly weak. Facing direction has only a modest effect (Figure 7).

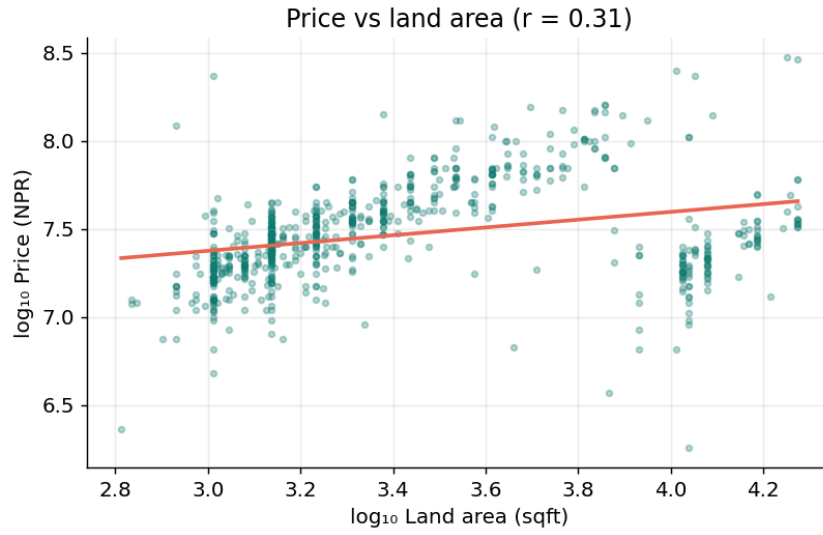


Figure 5: Price versus land area ($\log\text{-}\log$) with fitted trend line.

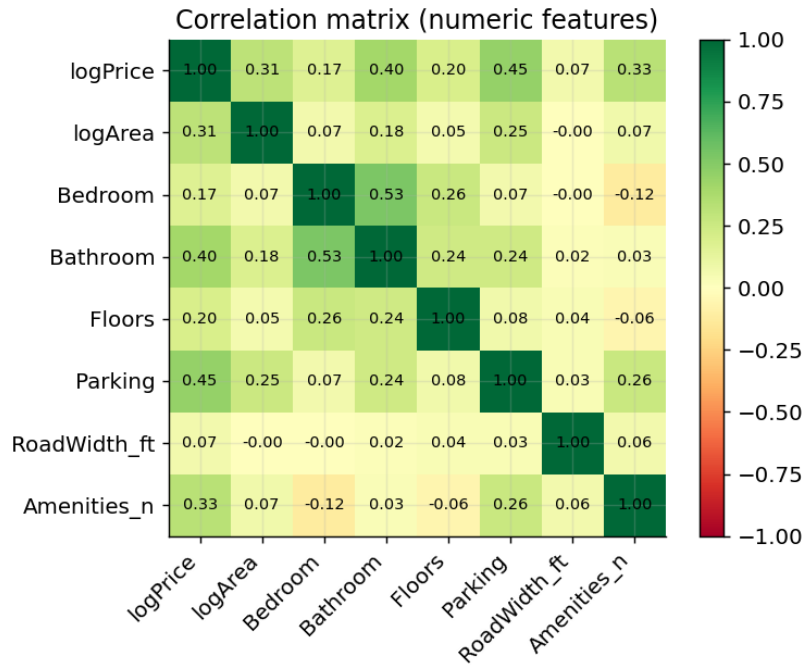


Figure 6: Correlation matrix of the numeric features.

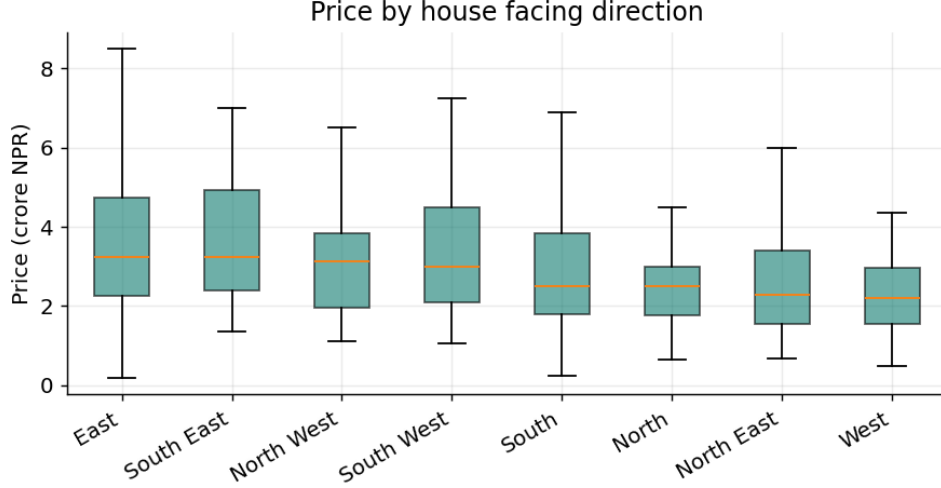


Figure 7: Price distribution by house-facing direction, differences are modest.

5 Methodology / Model

We framed the task as predicting $\log_{10}(\text{Price})$ from the cleaned features. The 738 houses were split 80/20 into a training set (590 houses) and a held-out test set (148 houses) using a fixed random seed for reproducibility. All evaluation metrics are reported on the test set only.

Features. Numeric inputs were $\log(\text{land area})$, bedrooms, bathrooms, floors, parking, road width and amenity count, each standardised to zero mean and unit variance. Categorical inputs, city, facing direction and road type, were one-hot encoded. The full preprocessing and model fitting were wrapped in a single `scikit-learn` pipeline (Pedregosa et al., 2011) to avoid data leakage. Data handling used `pandas` (McKinney, 2010) and `NumPy` (Harris et al., 2020) figures were produced with `matplotlib` (Hunter, 2007).

Models. We trained two models so we could compare a simple, interpretable baseline against a more flexible one:

- **Linear Regression**, an interpretable baseline that assumes price is a weighted sum of the features.
- **Random Forest Regressor** (300 trees, max depth 14), an ensemble of decision trees that captures non-linear effects and interactions between features (Breiman, 2001).

For context we also include a naïve baseline that simply predicts the training-set median price for every house. Performance is measured with the coefficient of determination (R^2), the root-mean-square error on the log scale, and the median absolute percentage error (MedAPE), which is the most intuitive: “typically, how far off is the prediction, in percent?”

6 Results

Both trained models clearly beat the naïve median baseline, and the Random Forest substantially outperforms Linear Regression (Table 3). The Random Forest explains 64% of the variance in (log) price and its typical prediction is within about 13% of the true listed price.

Table 3: Model performance on the held-out test set (148 houses).

Model	R^2 (test)	Median % error	Mean abs. error
Naïve median baseline	-0.03	,	—
Linear Regression	0.43	26.8%	NPR 1.46 crore
Random Forest	0.64	13.3%	NPR 0.99 crore

The predicted-versus-actual plot for the Random Forest (Figure 8) shows points clustering tightly around the diagonal, confirming a good fit across the price range. The feature-importance chart (Figure 9) quantifies the EDA intuition: land area is by far the dominant driver, followed by parking capacity and bathroom count, both of which act as proxies for the overall size and quality of a property. Bedroom count, by contrast, contributes little, likely because it is inconsistently reported in the listings. The residuals (Figure 10) are centred on zero with no strong pattern.

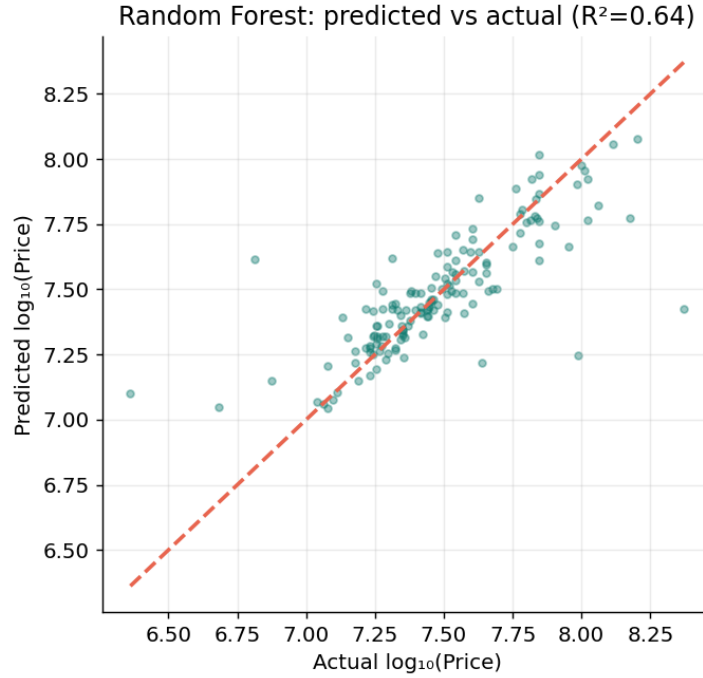


Figure 8: Random Forest predicted vs. actual log-price on the test set.

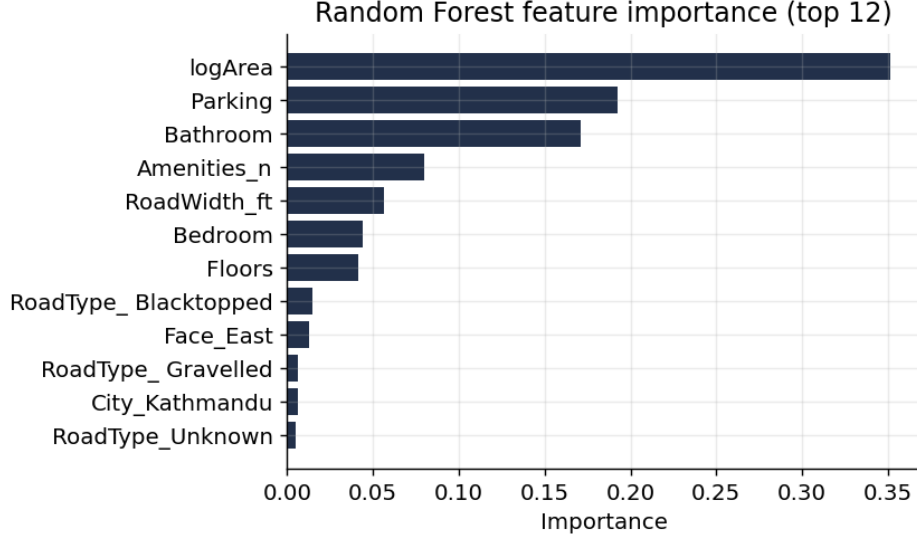


Figure 9: Random Forest feature importances, land area dominates.

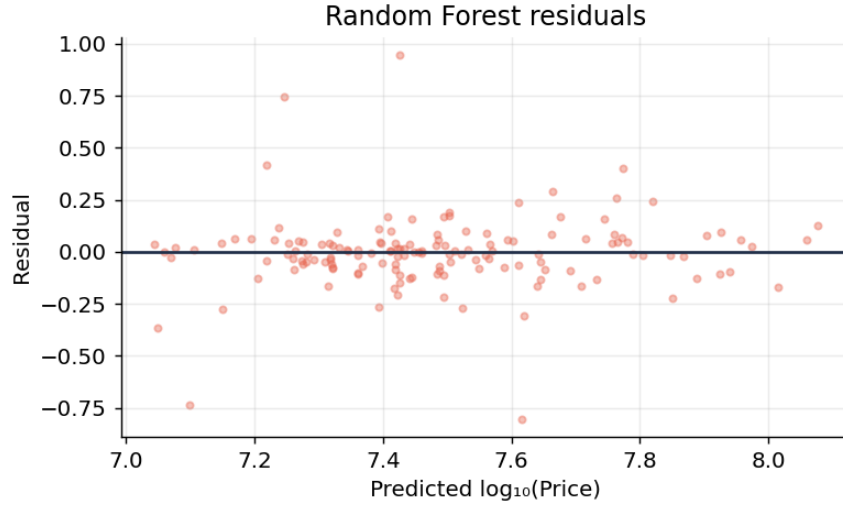


Figure 10: Residuals are centred on zero with no strong pattern.

7 Failure Analysis

Understanding what did not work is as informative as the headline results.

- **Linear Regression underfits.** With $R^2 = 0.43$ and a 27% typical error, the linear model is clearly too simple. Price depends on its features non-linearly and through interactions (e.g. a large plot is worth far more in central Kathmandu than on the valley edge), which a straight-line model cannot capture, hence the large gap to the Random Forest.
- **Severe data-quality loss.** We discarded two-thirds of the raw data (2,211 $\rightarrow$ 738) to obtain a clean sample. Junk prices, unparseable area strings and bare-land listings all had to be removed. The model is therefore only valid for the well-behaved subset and may not generalise to unusual listings.
- **Listed price $\neq$ sold price.** The target is the seller's asking price, not the actual transaction price. Nepali property typically sells below asking after negotiation, so the model learns aspirational prices, not market-clearing ones.

- **Missing location granularity.** We use only the city (3 valley districts). Within Kathmandu, neighbourhood matters enormously, but the free-text address field was too inconsistent to encode reliably, so a major source of price variation is unmodelled, a likely cause of the remaining residual scatter in Figure 8.
- **A single-day snapshot.** The data is one scrape from April 2020, so the model captures no time trend and cannot account for market movement or the effect of the COVID-19 period that immediately followed.

8 Future Improvements

- Geocode the address field to neighbourhood level (or attach latitude/longitude) to capture within-city location, the largest missing predictor.
- Try gradient-boosted trees (XGBoost / LightGBM) and tune hyper-parameters with cross-validation to push accuracy further.
- Engineer richer features from the amenities list (e.g. presence of a lift, modular kitchen or solar) rather than only counting them.
- Collect listings across multiple dates to model price trends over time.
- Where possible, validate against actual transaction prices to correct the asking-price bias.

9 Conclusion

We set out to predict Kathmandu Valley house prices from listing characteristics and to identify what drives them. After substantial cleaning of a noisy real-world dataset, reconciling nine land-measurement units, removing junk prices and scoping to genuine houses, we built a Random Forest model that explains 64% of price variation with a typical error of around 13%, comfortably outperforming both a linear model and a naïve baseline. The analysis confirms that land area is the dominant price driver, with parking and bathroom count acting as proxies for size and quality, while bedroom count and facing direction matter surprisingly little. In line with the spirit of the capstone, the value of this project lies less in raw accuracy than in a careful, well-documented pipeline and an honest account of its limitations, chiefly the gap between listed and sold prices and the absence of fine-grained location data.

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